

Basic Set Theory

Reference: Kenneth H. Rosen, *Discrete Mathematics and Its Applications*

1. Definition of a Set

A **set** is a collection of well-defined and distinct objects, considered as an object in itself. The objects in a set are called *elements* or *members* of the set.

Example: Let $A = \{1, 2, 3, 4\}$. Here $1 \in A$ but $5 \notin A$.

2. Representation of Sets

Sets can be represented in two main ways:

- *Roster form:* Listing all elements, e.g., $A = \{1, 2, 3\}$.
- *Set-builder form:* Describing the property of elements, e.g., $A = \{x \mid x \text{ is a natural number less than } 4\}$.

Example: Both forms represent the same set.

3. Types of Sets

- **Empty Set (Null Set):** A set containing no elements, denoted by $\emptyset$ or $\{\}$. **Example:** $A = \emptyset$
- **Singleton Set:** A set with exactly one element. **Example:** $B = \{5\}$
- **Finite Set:** A set with a countable number of elements. **Example:** $C = \{1, 2, 3, 4\}$
- **Infinite Set:** A set with infinitely many elements. **Example:** $D = \mathbb{N} = \{1, 2, 3, \dots\}$
- **Equal Sets:** Two sets A and B are equal if they have exactly the same elements. **Example:** $\{1, 2, 3\} = \{3, 2, 1\}$
- **Equivalent Sets:** Sets having the same number of elements (same cardinality). **Example:** $A = \{1, 2, 3\}, B = \{a, b, c\}$

4. Subsets

A set A is a **subset** of a set B , written $A \subseteq B$, if every element of A is also an element of B .

Example: If $A = \{1, 2\}$ and $B = \{1, 2, 3\}$, then $A \subseteq B$.

5. Proper Subset

A set A is a **proper subset** of B , written $A \subset B$, if $A \subseteq B$ but $A \neq B$.

Example: $\{1, 2\} \subset \{1, 2, 3\}$

6. Superset

A set B is a **superset** of A , written $B \supseteq A$, if $A \subseteq B$.

Example: $\{1, 2, 3\} \supseteq \{1, 2\}$

7. Power Set

The **power set** of a set A , denoted $P(A)$, is the set of all subsets of A , including the empty set and A itself.

Example: If $A = \{1, 2\}$, then $P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$

8. Universal Set

A **universal set**, denoted by U , is the set that contains all objects under consideration for a particular discussion.

Example: If $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 2\}$, then the complement of A is $A' = \{3, 4, 5\}$.

9. Cardinality of a Set

The **cardinality** of a set A , denoted $|A|$, is the number of elements in A .

Example: If $A = \{a, b, c\}$, then $|A| = 3$.

10. Set Operations

Let A and B be sets. Common set operations include:

- Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- Intersection: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- Difference: $A - B = \{x \mid x \in A \text{ and } x \notin B\}$
- Complement: $A' = \{x \mid x \notin A\}$ (with respect to universal set U)

Example: If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$, then $A \cup B = \{1, 2, 3, 4\}$, $A \cap B = \{3\}$, $A - B = \{1, 2\}$.

11. Cartesian Product

The **Cartesian product** of two sets A and B , denoted $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

Example: If $A = \{1, 2\}$ and $B = \{x, y\}$, then $A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$

12. Relations

A **relation** R from set A to set B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we say that a is related to b by R .

Example: Let $A = \{1, 2, 3\}$, $B = \{x, y\}$, and $R = \{(1, x), (2, y)\}$. Then R is a relation from A to B .

13. Types of Relations

Relations can have properties such as **reflexivity**, **symmetry**, and **transitivity**.

Example: On $A = \{1, 2, 3\}$, the relation $R = \{(1, 1), (2, 2), (3, 3)\}$ is reflexive, symmetric, and transitive — an **equivalence relation**.